REP-033 · Rules version 4 built: open stalls, the bell, Market Days, specialities and Market Gossip

Game	Fruit Market (app now on rules version 4, "tabletop edition")
Report date	2026-10-27 15:37 (Asia/Kuala_Lumpur)
Type	Build and validation report
Plans	PLAN-015 (approved: "approve plan 15") built together with the approved PLAN-014 items
Branch	`workflow/plan-015`, merged to `main` for the preview. **Live game unchanged**

Labels. [Sim] = computer players, 300 games per test, about ±5 points. **[Hyp]** = a guess about people.

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1. What was built

The day (five beats).

1. **Dawn:** the lorry unloads; the Tomorrow strip shows the next lorry, event, expected customer and the face-down walk-ins (fruit icons on the backs).
2. **Market:** buying, poorest shop first.
3. **Open stalls (new):** each shop secretly fills its for-sale crate; all crates show together.
4. **The bell:** one secret price per fruit, revealed together, with a short pause before the rush.
5. **The rush:** customers buy from the cheapest stall, leftover demand spills to the next; then evening and night.

Prototype A (core market).

- Uniform freshness: −$1 okay, −$2 not so, for every fruit.
- Ties: Favoured Trade, then Haggler, then fresher fruit, then poorer shop, then turn order.
- Cash catch-up kept ("poorest buys first").
- Options in the set-up: **Core market** (no traits, favours or specialities), **Street prices** (beginner) and **Market pulse** (advanced).

Prototype B (four specialities), drafted on the evenings after days 3, 6, 9 and 12: Cold Room $5, Juice Bar $4, Barrow $6, Premium Display $8. Shop Bell, Signboard, Bigger Shelf and Handcart are gone.

Prototype C (rhythm). Market Days on days 3, 6, 9 and 12 (+1 walk-in, no lorry); Last Call on the final day (+$1, then the jam maker buys leftovers at $1). Set-up toggle "Market Days", on by default.

Market Gossip replaces Free Cash (2 copies, evening): secretly see tomorrow's walk-ins. The fronts appear only on your screen, with a 👂 badge.

PLAN-014 items.

- Printed farmer cards (2–4 per fruit for 3 shops; 2 shops −1, 4 shops +1; missing fruit 1).
- Walk-ins: players − 2, then players − 1 in the second half. Customer cards are no longer scaled.
- Price shifts: blueberry −1, apple −2, mango +2.
- Events: Blueberry Glut and Apple Blight replace Slow Season and Rainy Day; Good Harvest +1 per 2 shops; Bad Harvest rounds up; no Health Inspector on the final day.
- `RULES_VERSION` 4. Older apps are asked to update; daily scores are kept separately per rules version.

App feedback.

- Stall screen and price screen: each shows rivals' crates, which customers can pay your price, and a hint.
- Day ledger: sales, bought, other (fines, favours), net, and fruit spoiling tonight; jam maker money on the results.
- Missed-sale reason "too old for $X". Bell pause before the rush. New tutorial tips.
- `RULES.md` and the in-app How to play rewritten.

One rule detail added for privacy. The evening is now held whenever anyone has a favour card. Before, it was held only when someone had an *evening* card, which told rivals what that player held. No balance effect.

2. Privacy (online)

Hidden until everyone commits	How it is protected	Check
Crates during Open stalls	Server keeps them aside; every view shows all crates empty	Unit test + browser test: PASS
Prices during the bell	Same	Unit test + browser test: PASS (0 leaks in 94 frames)
Face-down walk-ins	Each player's view shows only the fruit, no name, amount or price, unless that player heard the gossip	Unit test (fronts only to the gossiper) + browser test frame check: PASS
Rivals' favour cards	Only the number of cards is sent	Existing test: PASS

3. Validation

Check	Result	Evidence
`npm run verify` (records, lint, types, 92 unit tests, build)	PASS	Local run, commit after this report
`npm run verify:full` (solo game, performance budget, online match with leaver, quick match, 4 players, reconnect)	PASS	Exit 0; "leaks: 0"; "E2E OK"
New screens on a phone-sized browser	PASS	Stall, price, Market Day notice, evening, day ledger checked by screenshot; no page errors

Check	Result	Evidence
Paper or preview playtest (PLAN-015 T3)	NOT RUN	Needs the owner

4. Balance of the production build [Sim]

Measure	Result	Target or earlier figure
Hard beats Normal (2 players)	56%	REP-032: 56% (earlier aim 65%)
Hard beats Easy (2 players)	63%	—
Normal beats Easy (2 players)	55%	—
Comeback from last at half-time (3 players)	31%	REP-032: 32%
Day-5 leader wins (3 players)	34%	lower is more open
Blowouts, margin $20+ (2 / 3 / 4 players)	34% / 27% / 13%	watch 2-player games
Traits win (3 players, fair 33%)	30–38% (Big Basket highest)	REP-028: 30–38%
Old "first buyer rotates" rule for comparison	comebacks 8%, blowouts 58%	catch-up still essential

Specialities (3 players, fair 33%).

- Rushing **Barrow** wins 29% and **Premium Display** 27%; skipping either costs a lot (23%, 26%).
- Rushing **Cold Room** wins only 13%, **Juice Bar** 17%. The computers buy Juice Bar rarely (0.14 per game).
- Never building anything wins 19%, so specialities are worth having.
- This matches REP-032's warning: Cold Room and Juice Bar change *what you do*, and the computers don't play those plans well yet, so these numbers undersell them. [Hyp] Check in the playtest whether people find them useful.

Market feel.

- 90% of computer sales are within $1 of the customer's top price, and deep price wars are rare (0.1% of sales). Spoilage is near zero.
- Supply is tight enough that the computers rarely need to undercut. [Hyp] People may still undercut; watch for it in the playtest. If not, the next lever is +1 supply per listed fruit (needs a new plan).
- Market Days are lean days (income dips on days 3, 6 and 9), as designed.

5. Risks and open items

1. **Cold Room and Juice Bar are weak for computers.** Possible retune: Cold Room $4, Juice Bar $3. Not changed without your approval.
2. **Little undercutting among computers.** The hook ("how low dare I go?") needs a human check.
3. **2-player blowouts at 34%.** Watch in playtests.
4. **Hard vs Normal 56%,** below the old 65% aim. The computers are fair practice, not experts.

5. **Online settings:** online rooms use the defaults (Market Days on, no Core, Street or Pulse options). The playtest toggles are for games on one phone.

6. Next steps

1. Preview playtest in the order of REP-032 section 8: Core market first, then specialities, then Market Days, then Market Gossip.
2. Decide on the speciality retune (section 5, item 1).
3. PREPARE RELEASE 1.2.0 when you want rules version 4 on the live game (separate approval to publish).